RAJALAKSHMI ENGINEERING COLLEGE

Akshaya S 251001013-2025-IT

A2

Started on	Tuesday, 29 September 2026, 9:00 AM
State	Finished
Completed on	Tuesday, 29 September 2026, 10:00 AM
Time taken	1 hour
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];
    int dp[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
        dp[i] = 1;
    }

    int maxLength = 1;

    // DP: dp[i] = longest non-decreasing subsequence ending at i
    for (int i = 1; i < n; i++) {
        for (int j = 0; j < i; j++) {
            if (a[j] <= a[i] && dp[j] + 1 > dp[i]) {
                dp[i] = dp[j] + 1;
            }
        }

        if (dp[i] > maxLength) {
            maxLength = dp[i];
        }
    }

    printf("%d\n", maxLength);

    return 0;
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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